

AI Chatbot Specification

■ Property Agents

AI chatbot specification for property agents, real estate agencies, and property management in Singapore.

Suggested Pricing

Component	Price
One-time setup fee	\$500 - \$800
Monthly maintenance	\$100 - \$150/month
Add-on: WhatsApp integration	+\$150-200
Add-on: Payment collection	+\$100-150
Add-on: Custom integrations	+\$100-200 each

Business Pain Points (What You Sell Against)

These are the problems the business owner already knows they have. Lead with these in your pitch — don't sell 'AI', sell the solution to their pain.

- Every listing gets the same 10 questions: price, size, floor, facing, lease, distance to MRT.
- Lead qualification is manual — agents waste time on unqualified enquiries.
- Juggling enquiries from PropertyGuru, Instagram, WhatsApp — messages get lost.
- Hot leads go cold because response time is too slow (competing agents respond faster).
- Scheduling viewings requires 5-10 messages back and forth to find a mutual time.
- Weekend open houses generate leads that aren't followed up on until Monday.

30-Second Sales Pitch

I'll build an AI assistant that answers listing questions instantly, qualifies leads by collecting budget and timeline, and books viewings automatically. You respond to hot leads only — the bot handles the tyre-kickers. Works on your website, WhatsApp, and PropertyGuru landing pages.

Feature Requirements

Build in order: Must Have first (your MVP), then Should Have (first upsell), then Nice to Have (premium tier).

Feature	Description	Priority	Effort
Listing FAQ	Answer questions about specific properties: price, size, floor plan, amenities	Must Have	Low
Lead Qualification	Collect budget, timeline, requirements and score leads	Must Have	Medium
Viewing Booking	Schedule property viewings with calendar sync	Must Have	Medium
Multi-Listing Search	Search across multiple listings by criteria: location, budget, bedrooms	Should Have	High
Mortgage Calculator	Quick monthly payment estimate based on price and down payment	Nice to Have	Medium
Document Collection	Collect required documents from buyers for loan pre-qualification	Nice to Have	Medium

Detailed Feature Specifications

Use these specifications when building with Claude Code. Each feature includes the user story, technical implementation notes, and acceptance criteria.

Feature: Listing FAQ

Priority: Must Have | **Effort:** Low

User Story: As a buyer/renter, I want instant answers about a property without waiting for the agent to reply.

Technical Notes: Upload listing details as structured data. Handle property-specific queries: PSF, floor area, number of bedrooms, facing, tenure, nearby MRT, school proximity. Support multiple active listings.

Acceptance Criteria:

- Answers property-specific questions accurately
- Handles HDB-specific terms: lease, TOP date, flat type
- Handles condo terms: PSF, maintenance fee, facilities
- Supports queries across multiple active listings

Feature: Lead Qualification

Priority: Must Have | **Effort:** Medium

User Story: As an agent, I want the bot to qualify leads so I only spend time on serious buyers.

Technical Notes: Conversational qualification: budget range, timeline (when looking to buy/rent), financing status (cash/loan/HDB loan), current housing situation. Score leads as hot (ready to buy, budget matches) / warm (interested but not urgent) / cold (browsing). Route hot leads to agent immediately via Telegram.

Acceptance Criteria:

- Collects budget, timeline, financing, and requirements
- Scores leads as hot/warm/cold based on criteria
- Hot leads trigger immediate Telegram alert to agent
- Lead details compiled in a structured format for CRM

Feature: Viewing Booking

Priority: Must Have | **Effort:** Medium

User Story: As a buyer, I want to book a viewing at a time that works for me without back-and-forth messaging.

Technical Notes: Show available viewing slots from agent's Google Calendar. Collect: name, phone, preferred date/time, property interested in. Create calendar event. Send confirmation with property address and agent contact.

Acceptance Criteria:

- Shows real agent availability
- Handles 'this weekend' or 'any Saturday' naturally
- Creates calendar event with property address
- Sends viewing confirmation with directions
- Reminder sent 2hr before viewing

Feature: Multi-Listing Search

Priority: Should Have | **Effort:** High

User Story: As a buyer, I want to tell the bot what I'm looking for and get matching properties from the agent's portfolio.

Technical Notes: Store all active listings in a structured database. Parse user criteria: location, budget range, bedrooms, property type. Return matching listings sorted by relevance. If no match, suggest closest alternatives.

Acceptance Criteria:

- Filters listings by budget, location, bedrooms, property type
- Returns top 3-5 matches with key details
- Suggests alternatives if no exact match
- Handles SG-specific areas: 'near Jurong', 'in the East', 'District 15'

Feature: Mortgage Calculator

Priority: Nice to Have | **Effort:** Medium

User Story: As a buyer, I want a quick estimate of monthly mortgage payments.

Technical Notes: Simple calculator: input price, down payment %, loan tenure, interest rate (default 3.5%). Output monthly payment, total interest, total cost. Include disclaimer: 'This is an estimate. Consult a mortgage advisor for exact figures.'

Acceptance Criteria:

- Calculates monthly payment correctly
- Uses reasonable default interest rate for SG (3.5%)
- Handles HDB loan vs bank loan rate difference
- Always includes disclaimer about consulting advisor

Feature: Document Collection

Priority: Nice to Have | **Effort:** Medium

User Story: As an agent, I want to collect necessary documents from serious buyers before the viewing.

Technical Notes: For serious leads: request IPA (In-Principle Approval) status, income documents, proof of funds. Guide buyer through what documents are needed based on purchase type (HDB vs condo, citizen vs PR vs foreigner). DO NOT collect actual documents via chatbot — just guide and checklist.

Acceptance Criteria:

- Provides correct document checklist based on buyer profile
- Differentiates HDB vs private purchase requirements
- Does NOT collect or store actual financial documents
- Sends checklist via WhatsApp for easy reference

Sample Conversations

These demonstrate expected chatbot behaviour. Use these as test cases during development.

Scenario: Buyer enquiry about a listing

Customer: Is the Clementi condo still available?

Bot: Yes, the 3BR at Clementi Peaks is still available. 1,100 sqft, asking \$1.35M. 5-min walk to Clementi MRT, 99-year lease from 2018. Would you like more details or schedule a viewing?

Customer: What's the monthly mortgage if I put 20% down

Bot: With 20% down (\$270K) on a 25-year bank loan at ~3.5%, your estimated monthly payment is about \$4,850. This is a rough estimate — I can connect you with a mortgage advisor for exact figures. Would you like to schedule a viewing this weekend?

Customer: Yes Saturday afternoon

Bot: I have 2pm and 4pm available this Saturday. Which works better for you?

Technical Architecture

Core Stack (Required)

Component	Technology	Purpose
LLM	Claude API (claude-sonnet-4-20250514)	Understands questions and generates answers from context
Vector Database	ChromaDB (local) or Pinecone (cloud)	Stores document embeddings for RAG retrieval
Embedding Model	Voyage AI or OpenAI text-embedding-3-small	Converts text chunks to vectors
Backend	Python FastAPI	API server handling chat requests
Frontend Widget	Vanilla JS embeddable snippet	Chat bubble on client's website
Admin Dashboard	Streamlit or Next.js	Where business owner uploads docs and views analytics

Add-on Integrations

- **Google Calendar API:** Viewing scheduling, agent availability management
- **WhatsApp Business API:** Primary communication channel for SG property market
- **Telegram Bot:** Instant hot lead alerts to agent
- **Google Sheets / Airtable:** Listing database, lead CRM
- **PropertyGuru API (if available):** Sync listings from PropertyGuru

Data Model (Documents to Collect from Client)

These are the documents and data you need from the business owner during onboarding. The quality of these documents directly determines chatbot accuracy.

- **Active Listings:** Spreadsheet: address, price, size, bedrooms, floor, facing, tenure, TOP date, nearest MRT, facilities, photos
- **Agent Profile:** Name, CEA registration, contact, areas of specialisation, languages spoken
- **Buyer Document Checklists:** Required docs by buyer type: citizen HDB, citizen condo, PR, foreigner
- **Common Property FAQs:** Stamp duty explanation, ABSD rates, HDB eligibility, CPF usage rules
- **Viewing Preparation Guide:** What buyers should check during viewing, questions to ask

PDPA Compliance Checklist (Singapore)

- Display clear data collection notice before collecting any personal data
- Get explicit consent before storing name, phone, email
- Provide opt-out option: 'Type STOP to opt out of data collection'
- Do not store NRIC, credit card numbers, or medical records in chat logs
- Implement data retention policy: auto-delete chat logs after 90 days
- Provide data export/deletion on request from the business owner
- Add privacy policy link in the chat widget footer

Singapore Localisation Notes

- Support Singlish phrasing: 'got discount ah?', 'how much one?', 'can book or not'
- Handle mixed language queries (English + Mandarin terms are common)
- Use SGD (\$) for all pricing — never USD
- Time format: 12-hour with AM/PM (e.g., 3pm, not 15:00)
- Support PayNow QR code generation for payments
- Reference local landmarks for directions ('near Jurong East MRT')
- Public holiday awareness for Singapore calendar